

E-Commerce Sales Analysis Using SQL

**Discovering Business Insights
from the
Olist E-Commerce Dataset**

Prepared By: Sidra Riaz

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Section 1: Executive Overview

1. Total Customers: 99441

Query: `select count(customer_id) from olist_customers_dataset;`

2. Total Unique Customers: 96096

Query: `select count(distinct(customer_unique_id) from olist_customers_dataset;`

3. Total Orders: 99441

Query: `select count(order_id) from olist_orders_dataset;`

4. Total Revenue: 16009004

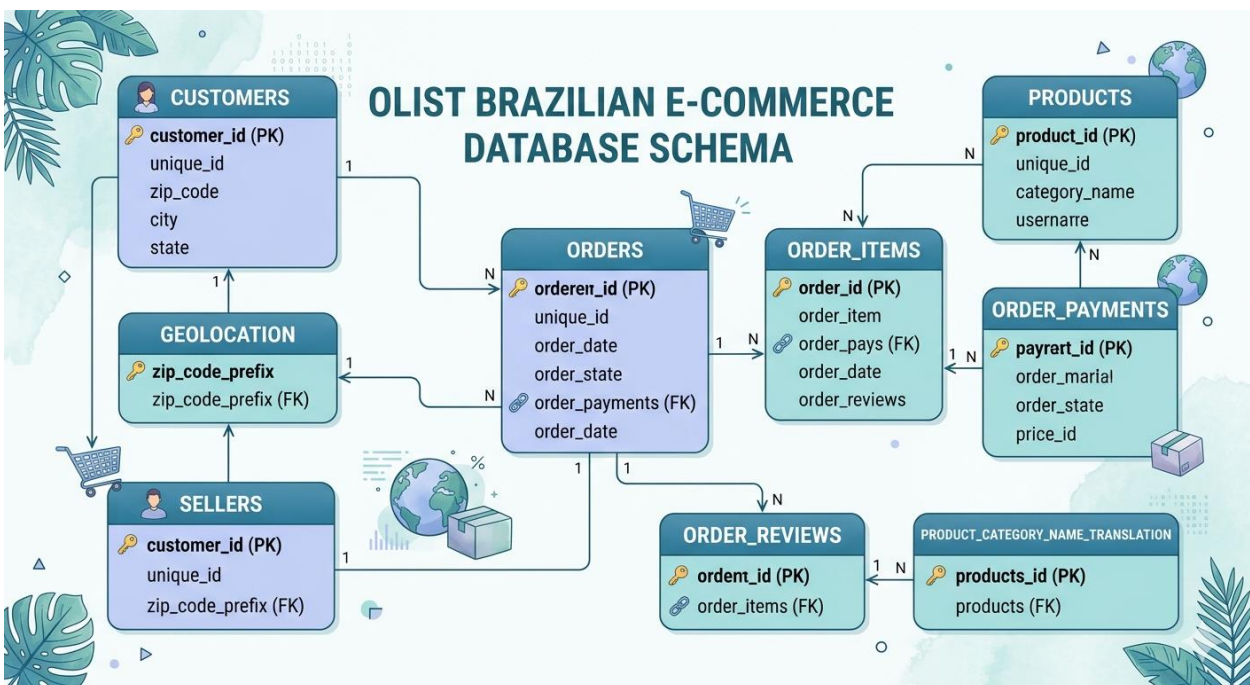
Query: `select sum(payment_value) from olist_order_payments_dataset;`

5. Average Order Value: 120.65

Query: `select avg(order_total) from (select order_id, sum(price) as order_total from olist_order_items_dataset group by order_id) t;`

6. Average Review Score: 4.08

Query: `select avg(review_score) from olist_order_reviews_dataset;`



Picture-1: The Database Schema

Section 2: Customer Insights

1. Which states have the most customers?

Query: `select customer_state, count(customer_id) as cos from olist_customers_dataset group by customer_state order by cos desc;`

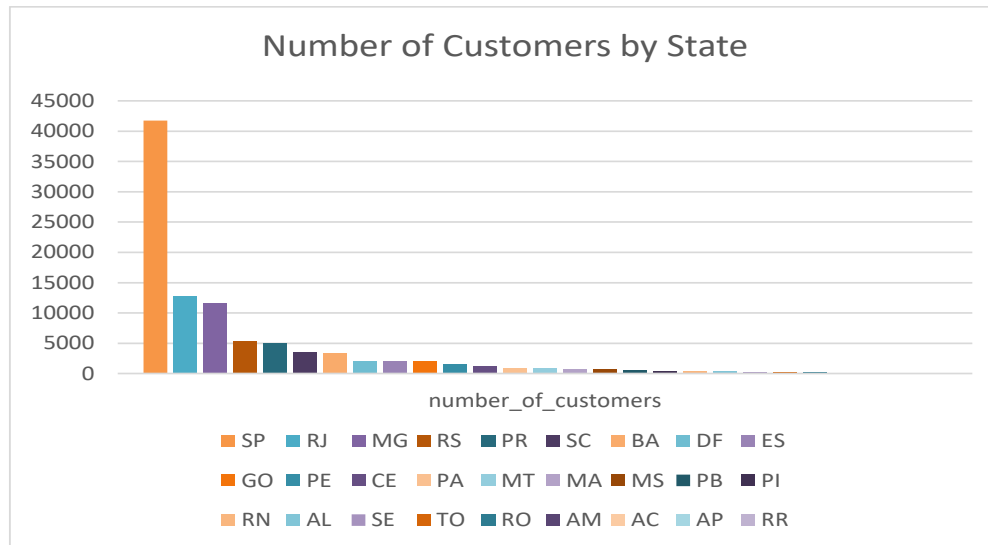


Figure-1

2. Which 10 cities have the most customers?

Query: `select customer_city, count(customer_id) as cos from olist_customers_dataset group by customer_city order by cos desc limit 10;`

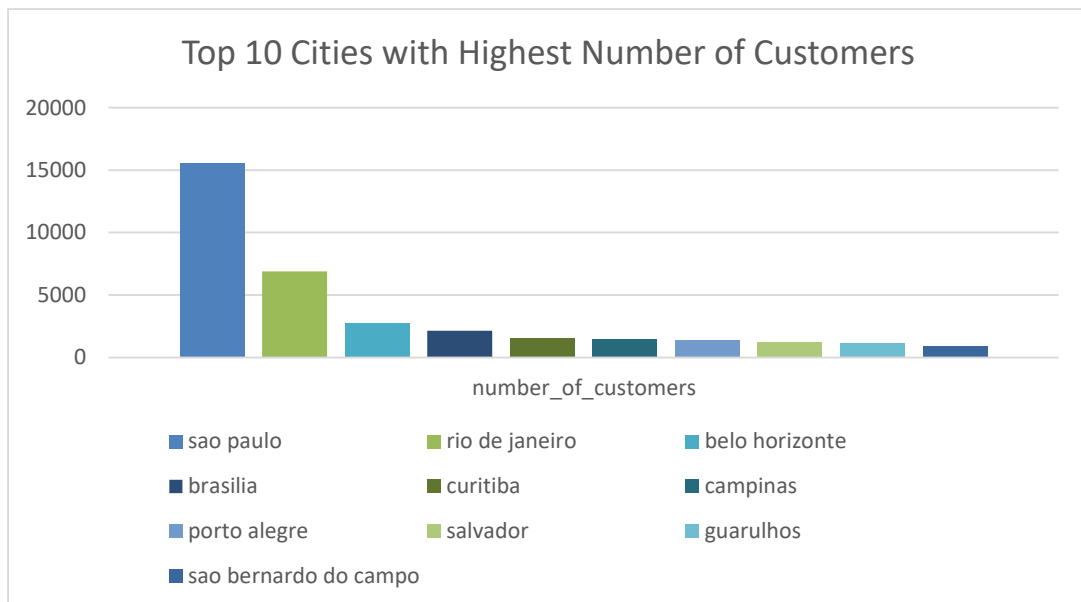


Figure-2

3. Which states generate the most revenue?

```
Query: select c.customer_state, sum(p.payment_value) as payt from
olist_customers_dataset c join olist_orders_dataset o on
c.customer_id=o.customer_id join olist_order_payments_dataset p on
o.order_id=p.order_id group by customer_state order by payt desc;
```

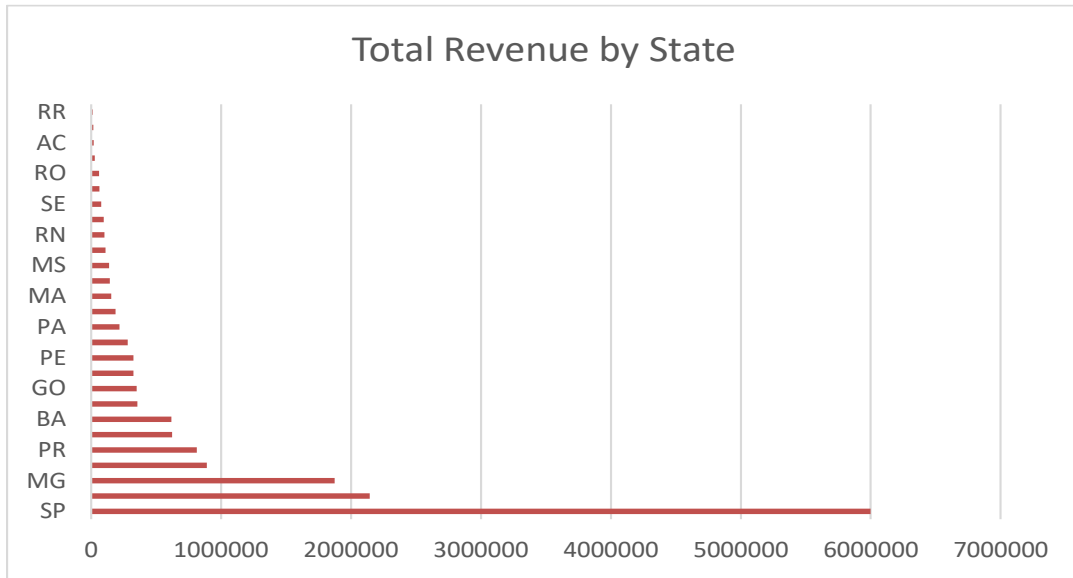


Figure-3

Section 3: Revenue Insights

1. What is the highest revenue generated according to payment type?

```
Query: select payment_type, sum(payment_value) as pv from
olist_order_payments_dataset group by payment_type order by pv desc;
```

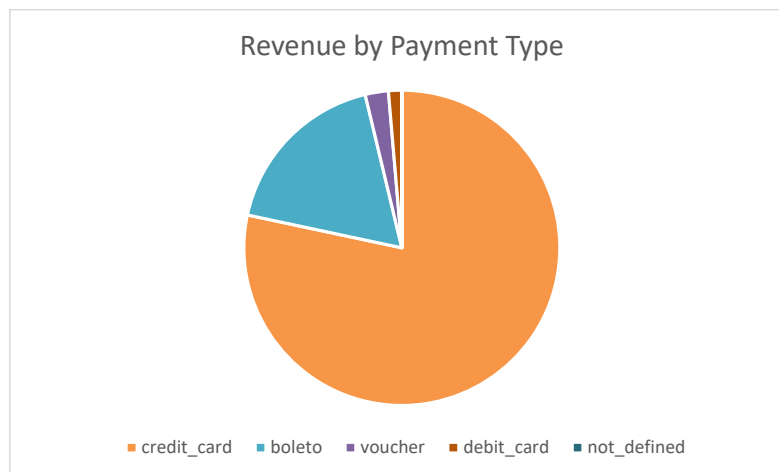


Figure-4

2. What are the top 10 cities that generated the most revenue?

```
Query: select c.customer_city, sum(p.payment_value) as payt from
olist_customers_dataset c join olist_orders_dataset o on
c.customer_id=o.customer_id join olist_order_payments_dataset p on
o.order_id=p.order_id group by customer_city order by payt desc limit
10;
```

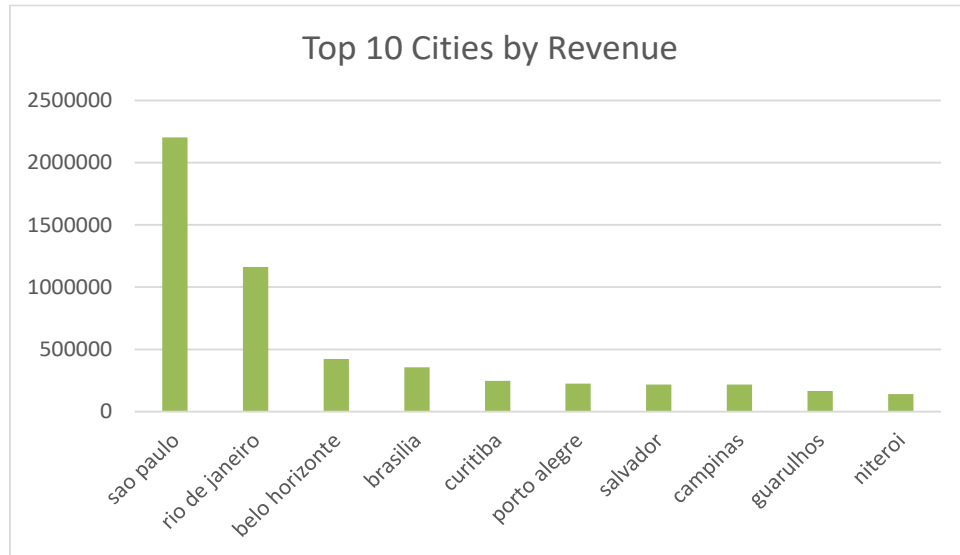


Figure-5

3. What is the average revenue generated by each state?

```
Query: select c.customer_state, avg(p.payment_value) as avg_revenue
from olist_customers_dataset c join olist_orders_dataset o on
c.customer_id = o.customer_id join olist_order_payments_dataset p on
o.order_id = p.order_id group by c.customer_state order by avg_revenue
desc;
```

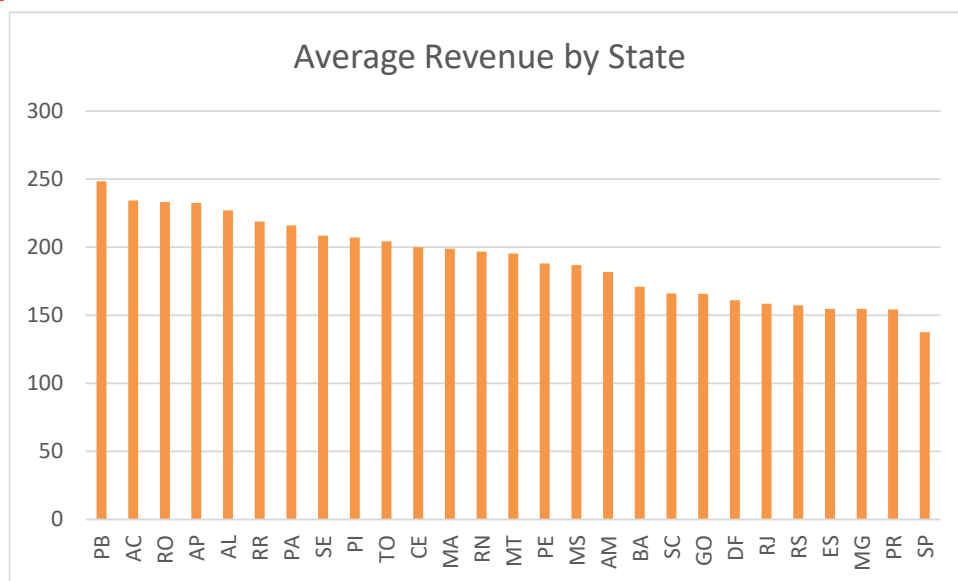


Figure-6

Section 4: Product Insights

1. What are the product categories that generate the highest revenue?

Query: `select product_category_name, sum(price) as p from olist_products_dataset pd join olist_order_items_dataset oi on pd.product_id=oi.product_id group by product_category_name order by p desc;`

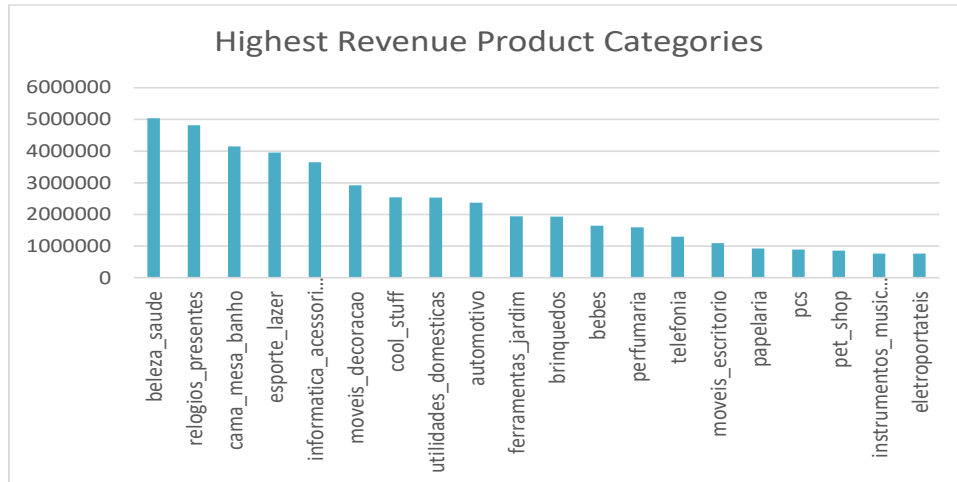


Figure-7

2. What are the categories with highest order count?

Query: `select count(order_id) as o, product_category_name from olist_order_items_dataset oi join olist_products_dataset op on oi.product_id=op.product_id group by product_category_name order by o desc;`



Figure-8

Section 5: Customer Satisfaction

1. Number of Total reviews: 99224

Query: `select count(review_id) from olist_order_reviews dataset;`

2. Percentage of positive reviews (4–5 stars): $= 76470/99224 \times 100 = 77.07\%$

Query: `select count(*) from olist_order_reviews dataset where review_score between 4 and 5;`

3. Percentage of negative reviews (1–2 stars): $= 14575/99224 \times 100 = 14.69\%$

Query: `select count(*) from olist_order_reviews dataset where review_score IN (1,2);`

4. Percentage of neutral reviews (3 stars): $8179/99224 = 8.24\%$

Query: `select count(*) from olist_order_reviews dataset where review_score = 3;`

5. What's the relationship between reviews and order status?

Query: `select o.order_status, round(avg(r.review_score), 2) as avg_review_score from olist_orders_dataset o join olist_order_reviews_dataset fixed r on o.order_id = r.order_id group by o.order_status order by avg_review_score desc;`

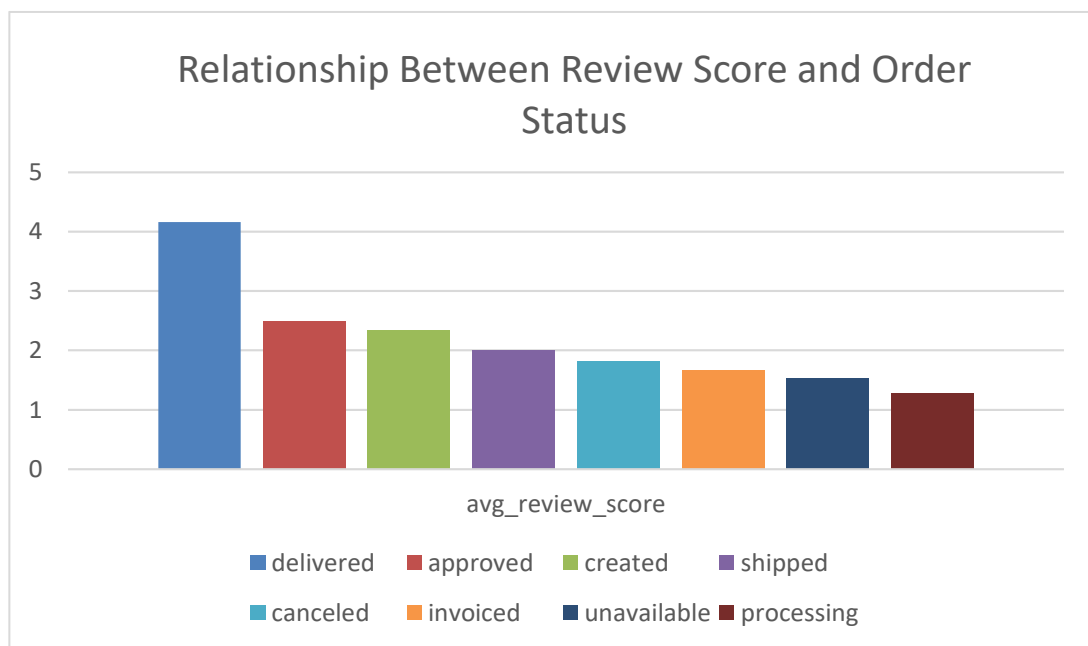


Figure-9

Section 6: Key Findings

1. Strong Customer Base with Repeat Purchases

The platform served **96,096 unique customers** and processed **99,441 orders**, indicating that a portion of customers returned to make additional purchases. This suggests a level of customer retention and repeat buying behavior.

2. Revenue Exceeded 16 Million

The business generated over **16 million in total revenue** from nearly 100,000 orders. With an average order value of approximately **120.65**, the platform demonstrates a healthy transaction volume and consistent customer spending.

3. São Paulo Dominates the Customer Base

The state of **São Paulo (SP)** recorded the highest number of customers, significantly outperforming other states. This indicates that SP is the company's largest market and a key contributor to overall business performance.

4. Major Cities Drive Customer Acquisition

Cities such as **São Paulo, Rio de Janeiro, Belo Horizonte, Brasília, and Curitiba** ranked among the highest in customer count. These urban centers represent important markets for future marketing and expansion efforts.

5. Revenue Is Concentrated in a Few States

States including **São Paulo, Minas Gerais, and Paraná** generated the highest revenue. This suggests that business performance is heavily concentrated in specific geographic regions.

6. Credit Cards Are the Preferred Payment Method

Among all payment methods, **credit cards generated the highest revenue**, making them the dominant payment option for customers. Businesses should continue optimizing card-based payment experiences to support sales growth.

7. Beauty and Health Products Generate the Highest Revenue

The **Beauty & Health (beleza_saude)** category produced the highest revenue among all product categories. Other strong-performing categories included watches, home goods, sports & leisure, and technology-related products.

8. Certain Product Categories Drive Most Orders

A relatively small group of product categories accounted for a large share of total orders, indicating that customer demand is concentrated in a few key product segments.

9. Customer Satisfaction Is Strong

The average review score was **4.08 out of 5**, indicating generally positive customer experiences across the platform.

10. Positive Reviews Significantly Outnumber Negative Reviews

Approximately **77.07%** of all reviews were positive (4–5 stars), while only **14.69%** were negative (1–2 stars). This suggests that most customers were satisfied with their purchases and overall shopping experience.

11. Neutral Reviews Represent a Small Portion of Feedback

Only **8.24%** of reviews were neutral (3 stars), indicating that customers generally had either a positive or negative experience rather than remaining indifferent.

12. Delivered Orders Receive Better Customer Ratings

Analysis of review scores by order status showed that **delivered orders consistently received the highest ratings**, while canceled, unavailable, and problematic orders received lower scores. This highlights the importance of successful order fulfillment in maintaining customer satisfaction.

Section 7: Recommendations

1. Strengthen Presence in High-Performing States

Since **São Paulo, Minas Gerais, and Paraná** generate the largest share of customers and revenue, the company should continue investing in targeted marketing campaigns and customer retention programs in these regions to maximize returns.

2. Expand Successful Product Categories

High-revenue categories such as **Beauty & Health** and other top-performing segments should receive greater inventory support, promotional focus, and product variety to capitalize on existing customer demand.

3. Improve Customer Retention

Although the platform recorded repeat purchases, the difference between total orders and unique customers suggests there is room for improvement. Loyalty programs, personalized offers, and email marketing campaigns could encourage customers to make additional purchases.

4. Optimize the Credit Card Payment Experience

As credit cards account for the largest share of revenue, the business should ensure a seamless checkout experience, offer installment options when appropriate, and minimize payment friction to reduce cart abandonment.

5. Focus on Order Fulfillment and Delivery Performance

The analysis showed that delivered orders receive significantly higher review scores than orders with other statuses. Improving delivery speed, shipment tracking, and fulfillment reliability can further increase customer satisfaction and positive reviews.

6. Investigate Causes of Negative Reviews

Although overall satisfaction is high, approximately **15% of reviews are negative**. The company should analyze low-rated orders to identify recurring issues such as delivery delays, damaged products, inaccurate descriptions, or customer service concerns.

7. Increase Marketing in High-Potential Cities

Top-performing cities already contribute a substantial portion of customers and revenue. Expanding localized advertising campaigns and partnerships in these cities can drive additional growth while leveraging established demand.

8. Use Customer Feedback for Continuous Improvement

With more than **99,000 reviews available**, customer feedback represents a valuable source of business intelligence. Regular review analysis can help identify emerging issues, improve products, and enhance the overall customer experience.

Conclusion

The analysis revealed a strong e-commerce business with over **99,000 orders, 16 million in revenue, and high customer satisfaction levels**. Customer activity and revenue are concentrated in a few major states and cities, while a small number of product categories contribute significantly to overall sales. By focusing on customer retention, product optimization, delivery performance, and data-driven decision making, the company can further improve revenue growth and customer satisfaction.